

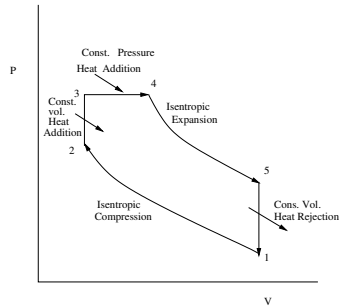
Dual Cycle

Processes

- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Volume Heat Addition
- 3 3-4 Cons. Pressure Heat Addition
- 4 4-5 Rev. Adiabatic Expansion
- 5 5-1 Const. Volume Heat Rejection

Dual cycle

P-V Diagram



Analysis

Heat supplied $Q_1 = mc_v(T_3 - T_2) + mc_p(T_4 - T_3)$

Heat rejected $Q_2 = mc_v(T_5 - T_1)$

Efficiency $\eta = 1 - \frac{Q_2}{Q_1}$

Compression ratio $r_k = \frac{V_1}{V_2}$

Expansion ratio $r_e = \frac{V_5}{V_4}$

Cut off ratio $r_c = \frac{V_4}{V_3}$

Constant volume pressure ratio $r_p = \frac{P_3}{P_2}$

$$\eta_{Dual} = 1 - \frac{r_p r_c^\gamma - 1}{r_k^{\gamma-1} r_p - 1 + \gamma r_p (r_c - 1)}$$

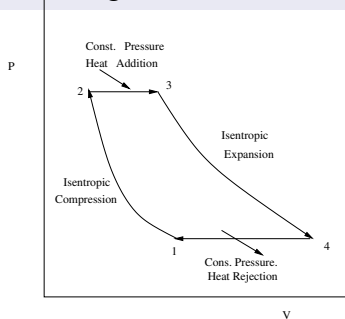
Brayton Cycle

Processes

- 1 1-2 Rev. Adiabatic Compression
- 2 2-3 Const. Pressure Heat Addition
- 3 3-4 Rev. Adiabatic Expansion
- 4 4-1 Const. Pressure Heat Rejection

Brayton cycle

P-V Diagram



Analysis

$$\text{Heat supplied } Q_1 = mc_p(T_3 - T_2)$$

$$\text{Heat rejected } Q_2 = mc_p(T_4 - T_1)$$

$$\text{Efficiency, } \eta = 1 - \frac{Q_2}{Q_1}$$

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$\frac{T_2}{T_1} = \left[\frac{p_2}{p_1} \right]^{\frac{\gamma-1}{\gamma}} = \frac{T_3}{T_4}$$

$$\frac{T_4}{T_1} - 1 = \frac{T_3}{T_2} - 1$$

$$\frac{T_4 - T_1}{T_3 - T_2} = \frac{T_1}{T_2} = \left[\frac{p_1}{p_2} \right]^{\frac{\gamma-1}{\gamma}} = \left[\frac{v_2}{v_1} \right]^{\gamma-1}$$

$$\eta_{\text{Brayton}} = 1 - \left[\frac{v_2}{v_1} \right]^{\gamma-1} = 1 - \frac{1}{r_k^{\gamma-1}} = 1 - \frac{1}{r_p^{\frac{\gamma-1}{\gamma}}}$$

$$r_k = \frac{v_1}{v_2}, r_p = \frac{p_2}{p_1}$$

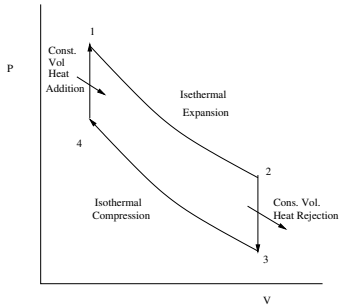
Stirling Cycle

Processes

- 1 1-2 Rev. Isothermal Expansion
- 2 2-3 Const. Volume Heat Rejection
- 3 3-4 Cons. Isothermal Compression
- 4 4-1 Const. Volume Heat Addition

Stirling cycle

P-V Diagram



$$Q_{1-2} = W_{1-2} = RT_1 \ln \frac{v_2}{v_1}$$

$$Q_{2-3} = -c_v(T_2 - T_1)$$

$$Q_{3-4} = W_{3-4} = -RT_2 \ln \frac{v_3}{v_4}$$

$$Q_{4-1} = c_v(T_1 - T_2)$$